

Linear Algebra HW 9

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Question 1

Let $\vec{w}_1 = \vec{v}_1$

$$\begin{aligned}\vec{w}_2 &= \vec{v}_2 - \frac{\langle \vec{w}_1, \vec{v}_2 \rangle}{\langle \vec{w}_1, \vec{w}_1 \rangle} \vec{w}_1 \\ &= \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{2}{6} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} \frac{2}{3} \\ \frac{2}{3} \\ \frac{2}{3} \end{bmatrix}\end{aligned}$$

$$\begin{aligned}\vec{w}_3 &= \vec{v}_3 - \frac{\langle \vec{w}_1, \vec{v}_3 \rangle}{\langle \vec{w}_1, \vec{w}_1 \rangle} \vec{w}_1 - \frac{\langle \vec{w}_2, \vec{v}_3 \rangle}{\langle \vec{w}_2, \vec{w}_2 \rangle} \vec{w}_2 \\ &= \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} - \frac{3}{6} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} - (0)(\dots) \\ &= \begin{bmatrix} -\frac{1}{2} \\ 0 \\ \frac{1}{2} \end{bmatrix}\end{aligned}$$

Question 2

Let the basis $\{1, x, x^2, x^3\}$ have entries $\vec{v}_i$, $1 \leq i \leq 4$

Let $\vec{w}_1 = \vec{v}_1$

Note: If $f(x) = x^i$, i odd, $\int_{-a}^a f(x) dx = 0$

$$\begin{aligned}\vec{w}_2 &= \vec{v}_2 - \frac{\langle \vec{w}_1, \vec{v}_2 \rangle}{\langle \vec{w}_1, \vec{w}_1 \rangle} \vec{w}_1 \\ &= x - \frac{\int_{-2}^2 x dx}{\int_{-2}^2 dx} \\ &= x\end{aligned}$$

$$\begin{aligned}\vec{w}_3 &= x^2 - \frac{\int_{-2}^2 x^2 dx}{\int_{-2}^2 dx} - \frac{\int_{-2}^2 x^3 dx}{\int_{-2}^2 x^2 dx} x \\ &= x^2 - \frac{\frac{16}{3}}{4} \\ &= x^2 - \frac{4}{3}\end{aligned}$$

$$\begin{aligned}
\vec{w}_4 &= x^3 - \frac{\int_{-2}^2 x^3 dx}{\int_{-2}^2 dx} - \frac{\int_{-2}^2 x^4 dx}{\int_{-2}^2 x^2 dx} x - \frac{\int_{-2}^2 (x^2 - \frac{4}{3})(x^3) dx}{\int_{-2}^2 x^4 dx} x^2 \\
&= x^3 - \frac{\frac{64}{5}}{\frac{16}{3}} x \\
&= x^3 - \frac{12}{5} x
\end{aligned}$$

Hence $\{\vec{w}_1, \vec{w}_2, \vec{w}_3, \vec{w}_4\}$ form an orthogonal basis for $\mathbb{P}_2[x]$ with the given inner product
To get an orthonormal basis, we simply divide by the norm of the vectors w.r.t. to the inner product

$$\begin{aligned}
\langle \vec{w}_1, \vec{w}_1 \rangle &= \int_{-2}^2 dx = 4 \\
\langle \vec{w}_2, \vec{w}_2 \rangle &= \int_{-2}^2 x^2 dx = \frac{16}{3} \\
\langle \vec{w}_3, \vec{w}_3 \rangle &= \int_{-2}^2 x^4 dx = \frac{64}{5} \\
\langle \vec{w}_4, \vec{w}_4 \rangle &= \int_{-2}^2 x^6 dx = \frac{256}{7}
\end{aligned}$$

Hence $\left\{ \frac{1}{4}, \frac{3x}{16}, \frac{5(x^2 - \frac{4}{3})}{64}, \frac{7(x^3 - \frac{12}{5}x)}{256} \right\}$, forms an orthonormal basis

Question 3

(a)

(i)

Let $A, B, C \in M_{2 \times 2}(\mathbb{R})$, $\lambda \in \mathbb{R}$

Checking for linearity in the first variable,

$$\begin{aligned}
\langle \lambda A + C, B \rangle &= \text{tr}((\lambda A + C)B) \\
&= \text{tr}(\lambda AB + CB) \\
&= \lambda \text{tr}(AB) + \text{tr}(CB)
\end{aligned}$$

Hence it is linear in the first variable

Checking for linearity in the second variable,

$$\begin{aligned}
\langle A, \lambda B + C \rangle &= \text{tr}(A(\lambda B + C)) \\
&= \text{tr}(\lambda AB + AC) \\
&= \lambda \text{tr}(AB) + \text{tr}(AC)
\end{aligned}$$

Hence it is linear in the second variable

Hence it is bilinear

(ii)

$$\langle A, B \rangle = \text{tr}(AB)$$

With A having ij entries a_{ij} , and similar for B , we get

$$\begin{aligned}
AB &= \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & \dots \\ \dots & a_{21}b_{12} + a_{22}b_{22} \end{bmatrix} \\
\Rightarrow \text{tr}(AB) &= a_{11}b_{11} + a_{12}b_{21} + a_{21}b_{12} + a_{22}b_{22}
\end{aligned}$$

From this we see that $\text{tr}(AB) = \text{tr}(BA)$

Then

$$\langle A, B \rangle = \langle B, A \rangle$$

Hence it is symmetric

(iii)

$$\langle A, A \rangle = \text{tr}(A^2)$$

This is not always greater or equal to zero

Consider $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$, $A^2 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

Hence quite clearly $\text{tr}(A^2) < 0$

(b)

(i)

Let $A, B, C \in M_{2 \times 2}(\mathbb{R})$, $\lambda \in \mathbb{R}$

Checking for linearity in the first variable,

$$\begin{aligned} \langle \lambda A + C, B \rangle &= \text{tr}((\lambda A + C)^T B) \\ &= \text{tr}(\lambda A^T B + C^T B) \\ &= \lambda \text{tr}(A^T B) + \text{tr}(C^T B) \end{aligned}$$

Hence it is linear in the first variable

Checking for linearity in the second variable,

$$\begin{aligned} \langle A, \lambda B + C \rangle &= \text{tr}(A^T (\lambda B + C)) \\ &= \text{tr}(\lambda A^T B + A^T C) \\ &= \lambda \text{tr}(A^T B) + \text{tr}(A^T C) \end{aligned}$$

Hence it is linear in the second variable

Hence it is bilinear

(ii)

$$\langle A, B \rangle = \text{tr}(A^T B)$$

With A having ij entries a_{ij} , and similar for B , we get

$$A^T B = \begin{bmatrix} a_{11}b_{11} + a_{21}b_{21} & \dots \\ \dots & a_{12}b_{12} + a_{22}b_{22} \end{bmatrix}$$

$$\Rightarrow \text{tr}(A^T B) = a_{11}b_{11} + a_{21}b_{21} + a_{12}b_{12} + a_{22}b_{22}$$

From this we see that $\text{tr}(A^T B) = \text{tr}(B^T A)$

Then

$$\langle A, B \rangle = \langle B, A \rangle$$

Hence it is symmetric

(iii)

$$\langle A, A \rangle = \text{tr}(A^T A)$$

With ij entries as before we get

$$A^T A = \begin{bmatrix} a_{11}^2 + a_{12}^2 & \dots \\ \dots & a_{21}^2 + a_{22}^2 \end{bmatrix}$$

Hence $\text{tr}(A^T A) \geq 0$, with equality $\iff a_{ij} = 0, \quad \forall 1 \leq i, j \leq 2$

(c)

(i)

Let $A, B, C \in M_{2 \times 2}(\mathbb{R})$, $\lambda \in \mathbb{R}$

Checking for linearity in the first variable

$$\langle \lambda A + C, B \rangle = \det(\lambda A + C + B)$$

Since the determinant does not preserve addition, this is not bilinear

(ii)

$$\langle A, B \rangle = \det(A + B) = \det(B + A) = \langle B, A \rangle$$

Hence it is symmetric

(iii)

$$\langle A, A \rangle = \det(A + A) = 2 \det(A)$$

But we know the determinant of a matrix can be negative

Thus it is not positive definite

Question 4

Since A is real symmetric matrix there exists an orthogonal matrix B s.t. $B^{-1}AB$ is diagonal

$$\begin{aligned}\chi_A(\lambda) &= (2 - \lambda)((3 - \lambda)^2 - 1) \\ &= (2 - \lambda)(\lambda^2 - 6\lambda + 8) \\ &= (2 - \lambda)^2(\lambda - 4)\end{aligned}$$

Thus there are two eigenvalues $\lambda = 2$ and $\lambda = 4$

$\mathcal{N}(A - 4I)$:

$$\begin{aligned}&\begin{bmatrix} -1 & 1 & 0 & | & 0 \\ 1 & -1 & 0 & | & 0 \\ 0 & 0 & 2 & | & 0 \end{bmatrix} \\ \Rightarrow &\begin{bmatrix} 1 & -1 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \\ \Rightarrow V(4) &= \left\{ \begin{bmatrix} t \\ t \\ 0 \end{bmatrix} \mid t \in \mathbb{R} \right\}\end{aligned}$$

$\mathcal{N}(A - 2I)$:

$$\begin{aligned}&\begin{bmatrix} 1 & 1 & 0 & | & 0 \\ 1 & 1 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \\ \Rightarrow &\begin{bmatrix} 1 & 1 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \\ \Rightarrow V(2) &= \left\{ \begin{bmatrix} t \\ -t \\ s \end{bmatrix} \mid t, s \in \mathbb{R} \right\}\end{aligned}$$

Since eigenvectors associated with different eigenvalues of a symmetric matrix are always orthogonal, we only need to be careful with the two eigenvectors associated with $\lambda = 2$

Choosing them to be $\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, with the eigenvector associated with $\lambda = 4$ being $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, we get

$$B = \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Question 5

The matrix A defines a bilinear form by $\langle \vec{x}, \vec{y} \rangle = \vec{x}^T A \vec{y}$, $\vec{x}, \vec{y} \in \mathbb{R}^3$
Let $\vec{x}$ have components x_1, x_2, x_3 , similarly for $\vec{y}$

$$\begin{aligned}\langle \vec{x}, \vec{y} \rangle &= \begin{bmatrix} x_1 a + x_2 + x_3 & x_1 + x_2 a + x_3 & x_1 + x_2 + x_3 a \end{bmatrix} \vec{y} \\ &= y_1(x_1 a + x_2 + x_3) + y_2(x_1 + x_2 a + x_3) + y_3(x_1 + x_2 + x_3 a)\end{aligned}$$

To check for postive defitiness, we compute $\langle \vec{x}, \vec{x} \rangle$

$$\begin{aligned}\langle \vec{x}, \vec{x} \rangle &= x_1(x_1 a + x_2 + x_3) + x_2(x_1 + x_2 a + x_3) + x_3(x_1 + x_2 + x_3 a) \\ &= a(x_1^2 + x_2^2 + x_3^2) + 2x_1 x_3 + 2x_1 x_2 + 2x_2 x_3\end{aligned}$$

Since $(x_1 + x_2 + x_3)^2 = x_1^2 + x_2^2 + x_3^2 + 2(x_1 x_2 + x_1 x_3 + x_2 x_3)$
i.e. $2(x_1 x_2 + x_1 x_3 + x_2 x_3) = (x_1 + x_2 + x_3)^2 - (x_1^2 + x_2^2 + x_3^2)$
we have

$$\langle \vec{x}, \vec{x} \rangle = (a - 1)(x_1^2 + x_2^2 + x_3^2) + (x_1 + x_2 + x_3)^2$$

Thus we see that this is greater than equal to zero for $a \geq 1$
Hence A is postive definite for $a \geq 1$